

LMM Calibration & Hedging of an Interest Rate Structured Note

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1 Introduction

On 15 February 2008, Bank XX issues a 10-year structured bond, paying quarterly coupons indexed to Euribor 3M with cap-and-spread features that vary across three time buckets (years 0–3, 3–6, and 6–10). In exchange for these coupons (plus an upfront $X\%$ at the Start Date), the bank receives Euribor 3M + 2% on the same schedule. We work in a single-curve interest rate framework and neglect counterparty risk throughout.

2 1.a: LMM Spot Volatility Calibration

EUR Caps/Floors - Implied Volatilities													
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	1.50	1.75	2.00	2.25	2.5	3.0	3.5	4.0	5.0	6.0	7.0	8.0	10.0
1Y	14.0	13.0	12.9	12.1	13.3	13.8	14.4	15.0	17.2	19.1	20.2	21.6	23.9
2Y	22.4	19.7	17.5	18.0	19.2	20.4	21.0	21.4	22.3	23.6	24.9	26.1	28.1
3Y	23.8	21.7	20.0	19.8	20.3	20.5	20.8	21.4	22.9	24.3	25.6	26.7	28.2
4Y	24.2	22.4	20.9	20.4	20.4	20.2	20.2	20.5	21.7	22.9	24.0	25.0	26.6
5Y	24.3	22.6	21.2	20.6	20.4	19.8	19.5	19.6	20.5	21.5	22.6	23.5	25.0
6Y	24.3	22.7	21.4	20.7	20.2	19.4	18.9	18.8	19.3	20.2	21.2	22.0	23.5
7Y	24.1	22.6	21.4	20.7	20.1	19.1	18.4	18.1	18.4	19.1	20.0	20.8	22.2
8Y	23.9	22.5	21.4	20.6	20.0	18.8	18.0	17.6	17.6	18.2	19.0	19.8	21.1
9Y	23.7	22.4	21.3	20.5	19.8	18.5	17.6	17.1	17.0	17.6	18.3	19.0	20.3
10Y	23.5	22.2	21.2	20.4	19.6	18.3	17.3	16.8	16.5	16.9	17.6	18.3	19.5
12Y	23.0	21.7	20.8	20.0	19.3	17.9	16.9	16.2	15.8	16.0	16.5	17.1	18.1
15Y	22.3	21.2	20.3	19.5	18.7	17.3	16.3	15.5	15.0	15.1	15.5	16.0	16.9
20Y	21.6	20.4	19.5	18.8	18.0	16.6	15.5	14.7	14.1	14.1	14.5	15.0	15.9

Figure 1: Market quoted flat Cap Volatilities

The market quotes a 13×13 surface of *flat* Black volatilities $\sigma_{T_m, K}^{\text{flat}}$ at maturities $T_m \in \{1, 2, \dots, 10, 12, 15, 20\}$ years and strikes $K \in \{1.50, 1.75, 2, \dots, 10\}$, (in %). By definition, the flat vol of a T_m -cap at strike K is the unique constant $\sigma_{T_m, K}^{\text{flat}}$ that, plugged into the Black-76 formula for *every* caplet of the cap, recovers the market cap price:

$$\text{Cap}_m(K) = \sum_{i=1}^{4T_m-1} \text{Caplet}_i(\sigma_{T_m, K}^{\text{flat}}), \quad (1)$$

where Cap_m has $4T_m - 1$ caplets resetting quarterly (the first 3M Euribor is fixed at t_0 and carries no optionality). Each caplet is priced as

$$\text{Caplet}_i(\sigma) = \delta_i B(t_0, T_{i+1}) [L_i(t_0) N(d_1) - K N(d_2)], \quad d_{1,2} = \frac{\ln(L_i(t_0)/K) \pm \frac{1}{2}\sigma^2(T_i - t_0)}{\sigma\sqrt{T_i - t_0}}, \quad (2)$$

with $L_i(t_0) = L(t_0; T_i, T_{i+1})$, δ_i the Act/360 year fraction of the period $[T_i, T_{i+1}]$, and $(T_i - t_0)$ the Act/365 year fraction from t_0 to the reset date.

The calibration aims to recover, from the flat-vol surface, a finer grid of *spot* vols $\hat{\Sigma}_i(K)$, one per caplet $i = 1, \dots, 79$, that prices each cap exactly when used caplet-by-caplet.

2.1 Bootstrap procedure

Calibration is performed independently for each strike column K , via a sequential *bucket bootstrap* along maturities. For a given K :

1. **1Y bucket** The 1Y cap has only three caplets (resetting at 3m, 6m, 9m), and Eq. (1) alone admits the trivial solution

$$\hat{\Sigma}_1(K) = \hat{\Sigma}_2(K) = \hat{\Sigma}_3(K) = \sigma_{1Y,K}^{\text{flat}}. \quad (3)$$

This sets the first *pillar* value $\sigma_{1Y}^*(K) := \sigma_{1Y,K}^{\text{flat}}$, located at the reset of the last caplet of the 1Y cap, $T_3^{\text{res}} \approx 9$ months.

2. **Subsequent buckets** ($T_m \rightarrow T_{m+1}$). Define the cap-difference target

$$\Delta C_K^{m \rightarrow m+1} := \text{Cap}_{m+1}(K, \sigma_{T_{m+1},K}^{\text{flat}}) - \text{Cap}_m(K, \sigma_{T_m,K}^{\text{flat}}). \quad (4)$$

This corresponds to the sum of all caplets in the bucket $i \in \{4T_m, \dots, 4T_{m+1} - 1\}$:

$$\Delta C_K^{m \rightarrow m+1} = \sum_{i \in \text{bucket}} \text{Caplet}_i(\hat{\Sigma}_i(K)). \quad (5)$$

This is one equation in (typically) four unknown spot vols, and must be supplemented by an interpolation rule.

3. **Linear interpolation between pillars.** We assume that, between two adjacent pillars T_m and T_{m+1} , the spot vol is a linear function of the reset time:

$$\hat{\Sigma}_i(K) = \sigma_{T_m}^*(K) + \frac{T_i^{\text{res}} - T_m^{\text{pillar}}}{T_{m+1}^{\text{pillar}} - T_m^{\text{pillar}}} (\sigma_{T_{m+1}}^*(K) - \sigma_{T_m}^*(K)), \quad (6)$$

where $\sigma_{T_m}^*(K)$ is the spot vol at pillar T_m and $T_m^{\text{pillar}} := T_{4T_m-1}^{\text{res}}$ is the reset of the last caplet of the T_m -cap. With this convention, the last caplet of each bucket exactly matches the new pillar by construction.

The procedure is then repeated for the next bucket, using $\sigma_{T_{m+1}}^*(K)$ as the new left pillar. After looping through all maturities and all 13 strikes, one obtains the full 79×13 matrix of LMM spot volatilities.

2.2 Calibration results

Table 1 reports the calibrated LMM *pillar* spot volatilities $\sigma_{T_m}^*(K)$ obtained at the 13 maturity pillars and 13 strikes.

Table 1: Calibrated LMM pillar spot volatilities $\sigma_{T_m}^*(K)$ (in %).

$T \backslash K$	1.50	1.75	2.00	2.25	2.50	3.00	3.50	4.00	5.00	6.00	7.00	8.00	10.00
1Y	14.00	13.00	12.90	12.10	13.30	13.80	14.40	15.00	17.20	19.10	20.20	21.60	23.90
2Y	24.56	21.83	19.38	20.78	22.73	26.19	28.33	28.32	24.28	24.30	25.31	26.37	28.26
3Y	23.74	22.07	20.98	20.35	20.06	17.45	16.27	17.56	22.70	24.71	26.04	27.05	28.20
4Y	24.70	23.17	21.84	21.32	20.79	20.92	20.71	19.80	18.97	20.00	21.12	22.33	24.72
5Y	24.23	22.63	21.46	20.70	20.18	17.52	15.83	16.04	18.09	19.34	20.93	21.84	23.22
6Y	24.34	22.99	21.96	21.06	19.34	18.36	17.35	16.33	15.50	16.79	17.76	18.70	20.89
7Y	23.33	22.02	21.08	20.49	19.99	17.53	15.48	14.59	15.73	16.13	17.24	18.43	19.69
8Y	23.42	22.33	21.58	20.08	19.34	17.12	16.07	15.40	13.87	14.99	15.77	16.81	18.36
9Y	22.90	21.91	20.52	20.06	18.50	16.48	14.48	13.37	14.45	15.52	16.12	16.51	18.22
10Y	22.69	21.06	20.88	19.79	18.41	17.21	15.48	15.52	13.31	12.61	14.01	15.34	16.26
12Y	21.44	20.16	18.92	18.11	18.30	15.73	15.10	12.46	13.72	14.36	13.88	13.99	14.99
15Y	20.89	20.39	19.55	18.53	16.15	15.20	13.56	13.72	11.82	11.64	12.88	13.71	14.78
20Y	20.51	18.19	16.95	16.61	16.96	14.63	13.11	11.32	11.83	12.07	12.13	12.67	13.95

3 1.b: Pricing the Upfront $X\%$

The structured swap exchanges, on quarterly dates, two cash flows:

- **Bank XX (Party A) pays:** Euribor 3M + 2.00%, on the Act/360 day count, on each payment date T_{i+1} ;
- **I.B. (Party B) pays:** an upfront $X\%$ of the principal at the Start Date, plus quarterly coupons c_i paid on the same dates.

3.1 Structured coupon definition

The first quarterly coupon ($i = 1$) is fixed at $c_1 = 4\%$. From the second quarter onward, the coupon is a function of the Euribor 3M and depends on three time buckets:

$$c_i = \begin{cases} L_i + 1.00\% & \text{if } L_i \leq 4.20\% \\ 4.50\% & \text{if } L_i > 4.20\% \end{cases} \quad \text{for } T_i \in (0, 3\text{ y}], \quad (7)$$

$$c_i = \begin{cases} L_i + 1.20\% & \text{if } L_i \leq 4.70\% \\ 4.90\% & \text{if } L_i > 4.70\% \end{cases} \quad \text{for } T_i \in (3, 6\text{ y}], \quad (8)$$

$$c_i = \begin{cases} L_i + 1.30\% & \text{if } L_i \leq 5.40\% \\ 5.60\% & \text{if } L_i > 5.40\% \end{cases} \quad \text{for } T_i \in (6, 10\text{ y}]. \quad (9)$$

We denote the bucket parameters by (a, K, b) , with a the floating spread, K the fixing threshold, and b the cap value, summarized in Table 2.

Table 2: Parameters of the structured coupon, by time bucket.

Bucket	Spread a	Threshold K	Cap value b
0–3 y	1.00%	4.20%	4.50%
3–6 y	1.20%	4.70%	4.90%
6–10 y	1.30%	5.40%	5.60%

3.2 Coupon decomposition into vanilla payoffs

Each piecewise coupon (7)–(9) can be rewritten in closed form as:

$$c_i = (a + L_i) - (K + a - b) \mathbf{1}_{\{L_i > K\}} - (L_i - K)^+. \quad (10)$$

The decomposition (10) expresses each coupon as the sum of three vanilla components, all priced in closed form:

- a *floating-plus-spread* payoff $a + L_i$, whose risk-neutral expectation under the T_{i+1} -forward measure is simply $a + L_i^{\text{fwd}}(t_0)$;
- a *digital caplet* struck at K ;
- a *plain-vanilla caplet* struck at K , priced via the standard Black formula.

3.3 NPV of the two legs and upfront equation

Equating the NPV of the cash flows received by Bank XX with the NPV of the cash flows it pays yields:

$$\begin{aligned} X\% + \delta(t_0, T_1) B(t_0, T_1) \cdot 4\% + \sum_{i=1}^{N-1} \delta_i \left\{ B(t_0, T_{i+1}) [a + L_i^{\text{fwd}}(t_0)] - (K + a - b) P_i^{\text{dig}} - P_i^{\text{cap}} \right\} \\ = \sum_{i=0}^{N-1} \delta_i B(t_0, T_{i+1}) [L_i^{\text{fwd}}(t_0) + 2\%], \end{aligned} \quad (11)$$

where $N = 40$ is the total number of quarterly payments (10 years).

The volatilities $\hat{\Sigma}_i(K)$ entering each digital and caplet are interpolated by cubic spline (to preserve the shape of the curve and differentiability) across strike from the calibrated 79×13 LMM spot-vol matrix of Section 2 and evaluated at the bucket strike $K \in \{4.20\%, 4.70\%, 5.40\%\}$.

Solving Eq. (11) for $X\%$ yields:

$$\boxed{X\% = 12.9841\%} \quad (12)$$

4 1.g: Digital Risk Correction

The digital component in Eq. (10) was priced above using the Black formula

$$P_i^{\text{dig}} = \delta_i B(t_0, T_{i+1}) N(d_2), \quad (13)$$

which implicitly assumes that the implied volatility σ does not depend on the strike K . In reality, the cap volatility surface exhibits a *smile*.

4.1 Smile-adjusted digital pricing

For each bucket caplet i , the corrected digital reads

$$P_i^{\text{dig, corr}} = \underbrace{\delta_i B(t_0, T_{i+1}) N(d_2)}_{\text{Black digital}} - \underbrace{\frac{\partial \sigma_i}{\partial K} \bigg|_K}_{\text{smile correction}} \cdot \mathcal{V}_i(K), \quad (14)$$

where the caplet vega is

$$\mathcal{V}_i(K) = \delta_i B(t_0, T_{i+1}) L_i \sqrt{T_i - t_0} \varphi(d_1), \quad (15)$$

with $\varphi(\cdot)$ the standard normal pdf, and $\partial \sigma_i / \partial K$ is the slope of the spot-vol smile at strike K , evaluated analytically from the cubic spline interpolant of row i of the LMM spot-vol matrix (using `fnder` on the piecewise-polynomial form returned by `spline`).

Replacing P_i^{dig} with $P_i^{\text{dig, corr}}$ in every bucket of Eq. (11) and solving for $X\%$ yields:

$$\boxed{X^{\text{corr}}\% = 12.7677\%}, \quad (16)$$

i.e. a reduction of approximately 22 bps relative to the uncorrected price.

5 1.c: Delta-Bucket Sensitivities

In this part of the exercise, we computed the DV01 sensitivities for the single buckets.

Each bucket corresponds to one of the market instruments used in the bootstrap procedure (deposits, futures and swaps). For each bucket j , a $1bp$ shock is applied to the corresponding market quote. The yield curve is then re-bootstrapped and the structured product is re-priced.

The Delta sensitivity is defined as the variation in the NPV of the structured bond due to the increase of these rates, which corresponds to:

$$\Delta X_j = X^{(j)} - X_0, \quad \text{DV01}_j = -(N \cdot \Delta X_j),$$

where $X^{(j)}$ is the upfront of the certificate obtained under new curve settings and $N = 50$ million € is the notional.

The minus sign appears because X is the fair upfront that makes the product value equal to zero: an increase in the fair upfront corresponds to an opposite variation in the product NPV.

Results and interpretation Table 3 reports the most relevant fine bucket sensitivities. We do not display the full bucket table, since several instruments have either negligible or exactly zero sensitivity. This is expected, as only the market quotes that affect the portion of the bootstrapped curve used in the pricing of the 10-year structured bond can generate a material DV01.

Bucket	ΔX (bps)	DV01 (€)
Deposit 3	0.0144	-71, 95
Future 1	0.1113	-556, 48
Future 4	-0.0998	499, 17
Swap 6	0.5999	-2.999, 50
Swap 10	5.6448	-28.224, 00

Table 3: Most relevant fine bucket Delta sensitivities.

The results show that the Delta exposure is not uniformly distributed across the curve. Most deposits have a negligible impact on the product value, with only a small sensitivity appearing around the short-end node represented by Deposit 3. More material sensitivities arise in the futures segment and in the swap nodes up to the product maturity. In particular, the 10-year swap bucket generates the largest sensitivity, reflecting the importance of the terminal part of the curve for the valuation of the structured coupons.

Long-dated swaps beyond the product maturity exhibit zero sensitivity, since they do not affect the discount factors and forward rates entering the cash-flow valuation.

6 1.d : Computing Total Vega

The total Vega of the structured bond was computed by applying a parallel shift of 1% (100 basis points) to the market flat volatility surface. Following the shift, the entire model was recalibrated to ensure that the LMM spot volatilities were consistent with the new flat volatility conditions.

The contract was subsequently re-priced under these shocked volatility parameters. The total Vega, representing the absolute sensitivity of the bond's NPV to changes in implied volatility, is defined as:

$$\mathcal{V}_{Total} = \frac{NPV(\sigma_{new}) - NPV(\sigma_{old})}{bump} \quad (17)$$

where $bump = 0.01$ (equivalent to a 1% increase in volatility).

Interpretation of Results The computed Vega value for the structured bond is **-0.287321**. This result indicates that for every one percentage point increase in the whole market's implied volatility structure, the value of the structured bond decreases by approximately **-140.000,00 €**. Given the long-term maturity (10 years) and the presence of embedded digital options and caps within the coupon structure, the bond exhibits a significant exposure to volatility.

7 1.e: Coarse-Grained Delta and Hedging with Swaps

To obtain a more interpretable representation of the interest rate risk, the fine bucket sensitivities are aggregated into three coarse-grained buckets:

$$\mathcal{B}_1 = [0, 2] \text{ years}, \quad \mathcal{B}_2 = (2, 6] \text{ years}, \quad \mathcal{B}_3 = (6, 10] \text{ years}.$$

For each market quote used in the bootstrap we compute its maturity t_i . Then we define three weighting functions:

$$w_1(t_i) = \begin{cases} 1, & t_i \leq 2, \\ \frac{6 - t_i}{6 - 2}, & 2 < t_i < 6, \\ 0, & t_i \geq 6, \end{cases}$$

$$w_2(t_i) = \begin{cases} 0, & t_i \leq 2, \\ \frac{t_i - 2}{6 - 2}, & 2 < t_i \leq 6, \\ \frac{10 - t_i}{10 - 6}, & 6 < t_i < 10, \\ 0, & t_i \geq 10, \end{cases}$$

$$w_3(t_i) = \begin{cases} 0, & t_i \leq 6, \\ \frac{t_i - 6}{10 - 6}, & 6 < t_i < 10, \\ 1, & t_i \geq 10. \end{cases}$$

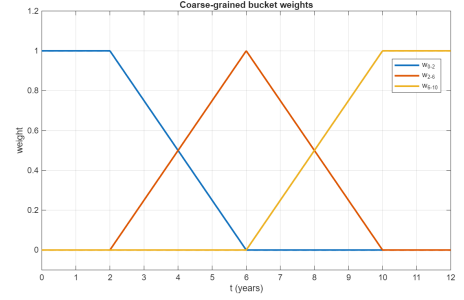


Figure 2: Triangular weighting functions used to construct the coarse-grained Delta shocks.

The j -th coarse-grained shock is therefore obtained by shifting every market quote i by

$$\Delta r_i^{(j)} = 1\text{bp} \cdot w_j(t_i).$$

For each of the three shock scenarios, we re-bootstrap the interest rate curve and reprice the structured product. The corresponding coarse-grained DV01 is then computed as the difference between the shocked and base NPV.

Coarse bucket	ΔX (bps)	DV01 (€)
0–2y	–0.2556	1.277, 80
2–6y	–0.1663	831, 25
6–10y	5.2116	–26.058, 00

Table 4: Coarse-grained Delta bucket sensitivities of the structured bond.

The results show that the dominant exposure is concentrated in the 6–10 year bucket, while the short and medium buckets have a much smaller impact.

Delta hedging strategy. The Delta risk is hedged using three payer interest rate swaps with maturities:

$$2\text{y}, \quad 6\text{y}, \quad 10\text{y}.$$

Each swap is priced at its par rate, so it has zero initial NPV. For each instrument, the sensitivity to the coarse buckets is computed using the same shock scenarios applied to the product, leading to a sensitivity matrix:

$$A \in \mathbb{R}^{3 \times 3}.$$

The hedge notionals n are obtained by solving:

$$\text{DV01}^{\text{prod}} + A \cdot n = 0.$$

The resulting hedge notionals are:

Maturity	Notional (€)	Position
2y	−6.800.000,00	Receiver
6y	−1.600.000,00	Receiver
10y	32.500.000,00	Payer

Table 5: IRS notionals used to hedge the coarse-grained Delta risk.

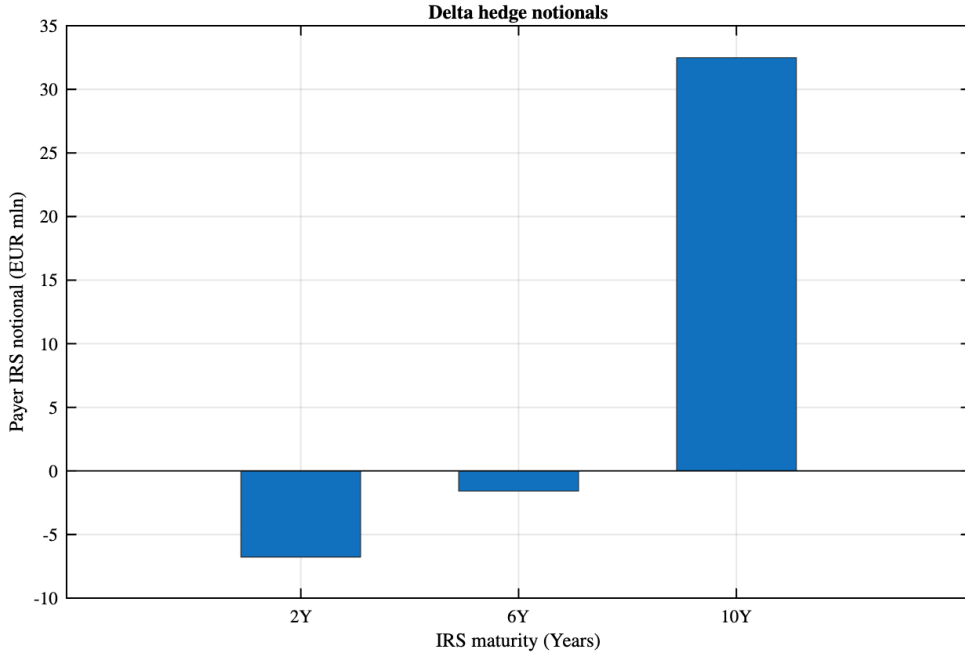


Figure 3: IRS notionals used to hedge the coarse-grained Delta risk

Under our sign convention, a positive notional corresponds to a payer IRS position, while a negative notional corresponds to the opposite receiver position. The hedge therefore requires receiver positions on the 2y and 6y swaps and a payer position on the 10y swap. This structure is consistent with the dominant long-end exposure observed in Table 4: the 10y payer IRS neutralizes the main 6–10y bucket risk, while the shorter receiver swaps correct the residual exposures generated on the front and intermediate parts of the curve.

Why start with the longest swap? The hedge is naturally interpreted by starting from the longest maturity. The 10y swap is the only hedge instrument with direct exposure to the 6–10y bucket, so it is first used to neutralize the long-end risk. The 6y and 2y swaps are then used to correct the residual exposures in the intermediate and short-end buckets.

The residual DV01 after hedging is numerically zero:

$$\text{DV01}^{\text{residual}} \approx 0,$$

confirming that the Delta risk is fully neutralized at the coarse-grained level.

8 1.f : Hedging via Caps

To neutralize the volatility risk across the curve, a bucketed Vega hedging strategy was implemented using two market instruments: a 6-year and a 10-year Cap, with strikes equal to the corresponding swap rate. The following methodology outlines the calibration and resolution of the hedging system.

Bucket Definition and Market Calibration The volatility shocks were structured into two buckets: 0-6 years and 6-10 years. For each bucket, a column vector of weights (bump matrix) was defined to accordingly distribute the 100 bps volatility shift across the LMM grid. The hedging strikes for the Caps were determined by taking the mid-price of the corresponding market swap rates ($K = \text{ATM Swap Rate}$). To determine the market price of each Cap, a spline interpolation was performed on the provided flat volatility surface to retrieve the implied volatility level corresponding to the selected ATM strike. This interpolated value was then utilized as the constant flat volatility input for the pricing model. By calibrating the baseline price in this manner, we ensure that the model perfectly mirrors the current market value of the hedging instruments.

Sensitivity Calculation and System Logic The Vega for each Cap was calculated by re-pricing the instrument under the shocked volatility scenarios:

$$\mathcal{V}_{Cap} = \frac{Price(\sigma_{bumped}) - Price(\sigma_{base})}{0.01} \quad (18)$$

The 6-year Cap, by construction, is only sensitive to the 0-6y bucket. Consequently, its sensitivity to the 6-10y shock is zero, creating an upper-triangular coefficient matrix. This mathematical structure allows the system to be solved numerically via back-substitution, starting with the 10Y Cap to isolate and hedge the tail risk (6-10y) before adjusting the front end (0-6y).

Linear System and Matrix Representation The hedging requirements are formalized in the following linear system $Ax = b$, where A represents the sensitivity matrix of the Caps and b represents the negative Vega of the structured bond:

$$\begin{bmatrix} \mathcal{V}_{6y}^{0-6} & \mathcal{V}_{10y}^{0-6} \\ 0 & \mathcal{V}_{10y}^{6-10} \end{bmatrix} \begin{bmatrix} N_{6y} \\ N_{10y} \end{bmatrix} = \begin{bmatrix} -\mathcal{V}_{Cert}^{0-6} \\ -\mathcal{V}_{Cert}^{6-10} \end{bmatrix} \quad (19)$$

Numerical Results The solution of the system yields the following notionals. As anticipated, a significant long position in the 10Y Cap is required to offset the digital risk in the second bucket, while a short position in the 6Y Cap is necessary to neutralize the resulting over-hedge in the first bucket, similarly to the DV01 results.

Hedging Instrument	Weight (Multiplier)	Cash Notional (EUR)
6Y ATM Cap	-1.5506	-77.530.000,00
10Y ATM Cap	1.7753	88.765.000,00

Table 6: Portfolio Notionals for Vega Hedging (Principal: 50M EUR).

8.1 Analysis of Results and Visual Interpretation

The relationship between the forward rate curve and the selected hedging strikes provides a clear narrative for the resulting notionals. By analyzing the market environment, we can anticipate the behavior of the linear solver before observing the final numerical outputs.

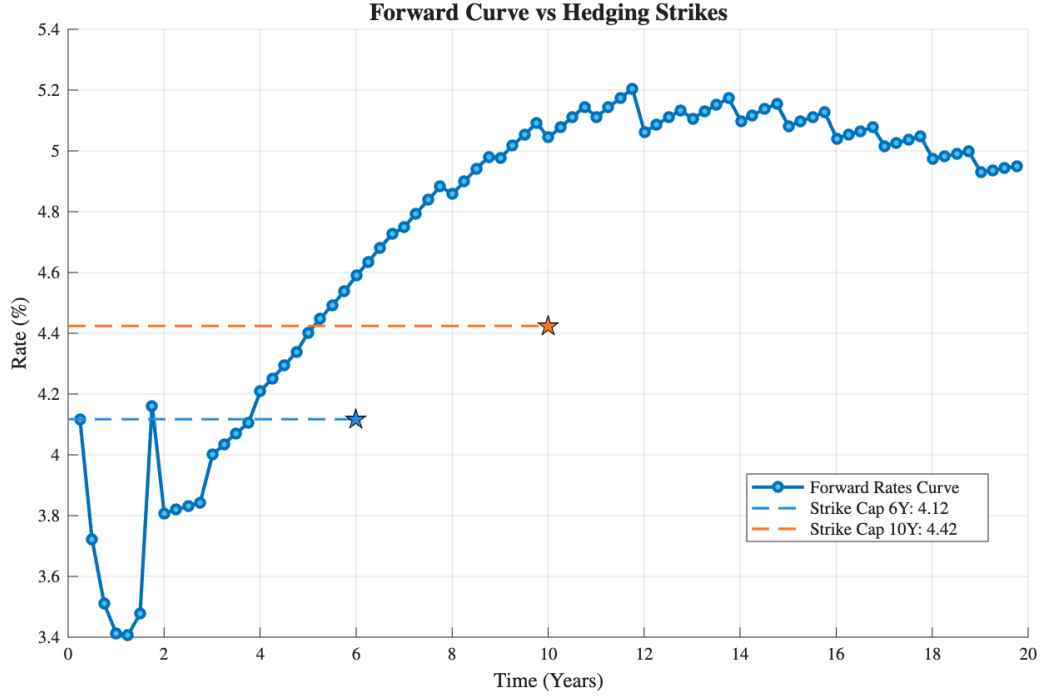


Figure 4: Forward Libor rate graph vs selected Caps strikes.

Vega of Cap Instruments As shown in Figure 3, the market environment exhibits a clear Vega misalignment:

- **Bucket 0-6 Years (Crossing Area):** The forward curve starts at 4.1%, dips to 3.4%, and subsequently rises sharply, crossing both the 6Y Cap strike (4.12%) and the 10Y Cap strike (4.42%) between years 3.5 and 5. In this region, uncertainty is maximized as the options are effectively At-The-Money (ATM), leading to extremely high Vega for the hedging instruments.
- **Bucket 6-10 Years (Deep ITM Area):** From year 6 onwards, the forward curve stabilizes between 4.6% and 5.2%, remaining consistently above the 10Y Cap strike (4.42%). Consequently, the caplets in this window are Deep In-The-Money (ITM). Since the payoff is nearly certain, the sensitivity to volatility (Vega) drops significantly.

Vega of Structured Bond The structured bond features a digital barrier (step-up coupon) set at 5.40% during the 6-10 year period. The forward curve in this segment hovers between 5.0% and 5.2%, placing the certificate's embedded digital option near the trigger level. This proximity results in a significant disparity between the bucketed sensitivities: the Vega for the first bucket (0-6y) is **-0.0203**, while the Vega for the second bucket (6-10y) reaches **-0.2508**. This confirms that the risk is heavily concentrated in the second part of the curve, where the certificate's exposure is more than twelve times higher due to the digital trigger proximity. To immunize this risk, the solver calculates:

$$N_{10y} \approx \frac{\text{Vega}_{Cert}^{6-10}}{\text{Vega}_{Cap10y}^{6-10}} \quad (20)$$

Because the 10Y Cap is Deep ITM in this region, its Vega (the denominator) is very small. To offset the certificate's high Vega (the numerator), the model requires an exceptionally high notional for the 10Y Cap. This situation forces a massive long position to compensate for the poor hedging efficiency of the instrument's strike.

In turn, this massive 10Y position generates a significant over-hedge in the 0-6Y bucket where the Cap is ATM. To maintain a net-zero Vega for the first bucket, the solver mandates a substantial short position in the 6Y Cap, leading to notionals far exceeding the 50M EUR principal. While mathematically sound, this is not an optimal hedge due to the high capital requirements and the elevated number of positions required to maintain the immunization.

Additional Hedging Scenarios To further test the robustness of the model, two additional scenarios were explored. First, a more granular 3-bucket approach was implemented by adding a 3Y Cap pillar: The positive notional for the 3Y Cap is consistent with the need to fine-tune the short-term hedge; however, this adds further complexity and capital commitment to the portfolio.

Instrument	Weight	Notional (EUR)
3Y Cap	0.5749	28.745.000,00
6Y Cap	-1.7863	-89.315.000,00
10Y Cap	1.7753	88.765.000,00

Table 7: 3-Bucket Granular Vega Hedge.

Instrument	Weight	Notional (EUR)
6Y Cap	-1.4000	-70.000.000,00
10Y Cap	1.7542	87.710.000,00

Table 8: Coarse-Grained Linear Bucket Results.

Finally, a "coarse-grained" bucketing approach was tested. Instead of rigid step-profile shifts, linear triangular weights were used to smooth the volatility bumps. This resulted in a softer and more efficient hedge: The reduction in the absolute weights suggests that smoother risk allocation reduces the extreme balancing acts required by the linear solver, though the leverage effect remains present due to the previous considerations.